

File: BalloonGame/BalloonConstants.py

```
# Constants used by more than one Python module

N_BALLOONS = 15 # number of balloons in a round of the game
BALLOON_MISSED = 'Missed' # balloon went off the top
BALLOON_MOVING = 'Balloon Moving' # balloon is moving
```

Listing 11-4: A module of constants that's imported by other modules

This is just a simple Python file that contains constants shared by more than one module. The main code needs to know how many balloons there are in order to display that number. The balloon manager needs to know the number so it can instantiate the correct number of `Balloon` objects. This approach makes it extremely simple to modify the number of `Balloon` objects. If we added levels with different numbers of balloons, we could build a list or a dictionary in this file alone, and all other files would have access to that information.

The other two constants are used in each `Balloon` object as status indicators as the balloon moves up the window. When I get to the discussion of the game play, you'll see that the balloon manager (`oBalloonMgr`) asks each `Balloon` object for its status, and each responds with one of these two constants. Placing shared constants in a module and importing that module in modules that use the constants is a simple and effective technique for ensuring that different parts of a program use consistent values. This is a good example of applying the Don't Repeat Yourself (DRY) principle by only defining values in a single place.

Main Program Code

The main code of our sample program, shown in [Listing 11-5](#), follows the 12-step template I've used throughout this book. It shows the user's score, the game status, and a Start button at the bottom of the window, and it reacts to the user clicking the Start button.